

Banasthali Vidyapith
Department of Computer Science
Course Handout: M. Tech. (CS/AI) II Semester
December 2025 – May 2026

Date: 19-December-2025

Course Code: CS 503

Course Name: Advanced Computer Architecture

Credit Points: 4

Max. Marks: 100 (CA: 40 + ESA: 60)

Course Instructor: Mr. Kanwaljeet Singh, Assistant Professor, Centre for Artificial Intelligence

Learning Outcomes: On successful completion of the course students will be able to:

- Understand the concepts of parallel processing and various computational models.
- Able to learn and evaluate the performance of multiprocessor systems using performance metrics.
- Design pipeline, superscalar and vector processing systems.
- Have the concepts of Multithreaded and dataflow parallel computing architectures.

Syllabus:

Section A

Overview of Modern Processor Architectures, Elements of Processor Design. Flynn's Taxonomy. Parallel Computer Models- Multiprocessors and Multi-computers. Principles of Scalable Performance- Performance Metrics and Measures. Memory Hierarchy- Cache and Cache Coherence, Bus Architecture.

Section B

Superscalar and Vector Processors. Multivector and SIMD Computers- Vector Processing Principles, Multivector Multiprocessors, SIMD Computer Organizations. Program and Network Properties: Conditions of Parallelism, Program Partitioning and Scheduling. Program Flow Mechanism- Control Flow Vs Data Flow. System Interconnects.

Section C

Principles of Pipeline Processing, Linear Pipeline Processors, Nonlinear Pipeline Processors- General Pipelines and Reservation Tables, Instruction Pipeline Design, Arithmetic Pipeline Design, Superscalar and Vector Pipeline Design, Scalable, Multithreaded and Dataflow Architectures.

Suggested Books:

- R1.** Hwang, K. (1993). Advanced Computer Architecture. McGraw-Hill Education.
- R2.** Sima D, Fountain, T., & Kacsuk, P. (2003). Advanced computer architectures: a design space approach. Pearson Education India.
- R3.** Patterson, D. A., & Hennessy, J. L. (2000). Computer Organization and Design. Morgan Kaufmann.
- R4.** Hwang, K., & Xu, Z. (1998). Scalable parallel computer architecture. McGraw-Hill.
- R5.** Hwang, K., & Faye, A. (1984). Computer architecture and parallel processing. McGraw Hill.
- R6.** El-Rewini, H., & Abd-El-Barr, M. (2005). Advanced computer architecture and parallel processing. John Wiley & Sons.

Suggested E-Learning Material:

- E1.** Advanced Computer Architecture
<https://www.iitg.ac.in/asahu/cs523e/>
- E2.** Computer Architecture
<https://www.coursera.org/learn/comparch>

Assessment Schedule:

Component	Marks	Submission/ Examination Date(s)	Allotment/ Syllabus
Assignment 1**	10	25 January, 2026	Topics shall be allotted in the class by 07 January, 2026
Periodical Test 1	10	01-05 February, 2026*	Lecture No. 01 to 22
Assignment 2**	10	04 March, 2026	Topics shall be allotted in the class by 11 February, 2026
Periodical Test 2	10	16-20 March, 2026*	Lecture No. 23 to 42
End-Semester Examination	60	12-30 April, 2026*	Lecture No. 01 to 50 (Entire Syllabus)

* Subject to change.

** Assignment evaluation will be based on written document(s)/ test(s)/ viva-voce/ any other component(s) as decided by the instructor(s) on regular basis.

Lecture-Wise Schedule:

Lecture Number	Topics to be Covered	Suggested Readings
1-2	Overview of Modern Processor Architectures	R1/R2
3-8	Elements of Processor Design. Flynn's Taxonomy. Parallel Computer Models- Multiprocessors and Multicomputers	R1/R2/R3/R4
9-12	Principles of Scalable Performance- Performance Metrics and Measures	R1/R2/R5
13-18	Memory Hierarchy- Cache and Cache Coherence, Bus Architecture	R1/R2
19-26	Superscalar and Vector Processors. Multivector and SIMD Computers- Vector Processing Principles, Multivector Multiprocessors	R1/R2/R3/R6
27-30	SIMD Computer Organizations. Program and Network Properties, Conditions of Parallelism	R1/R2
31-34	Program Partitioning and Scheduling. Program Flow Mechanism – Control Flow Vs Data Flow. System Interconnects.	R1/R2
35-39	Principles of Pipeline Processing, Linear Pipeline Processors, Nonlinear Pipeline Processors- General Pipelines and Reservation Tables	R1/R2/R3
40-43	Instruction Pipeline Design, Arithmetic Pipeline Design	R1/R2/R5
44-47	Superscalar and Vector Pipeline Design	R1/R2/R3/R6
48-50	Scalable, Multithreaded and Dataflow Architectures	R1/R2//R3

Kanwaljeet Singh